

Signal Processing for Machine Learning

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Introduction



Introduction

- Machine learning (ML) is the field of study that gives computers the ability to learn without being explicitly programmed
 - ML techniques are being successfully applied in countless areas of science and engineering
- Traditionally, ML approaches have been viewed as *black boxes* that process input signals to classify them or predict new values
 - Signals are everywhere in the physical world (the physical world is in fact a beacon of signals)
- However, input signals must be captured, processed, and manipulated before being presented to ML algorithms
 - Procedures that some deep learning practitioners consider non-essential



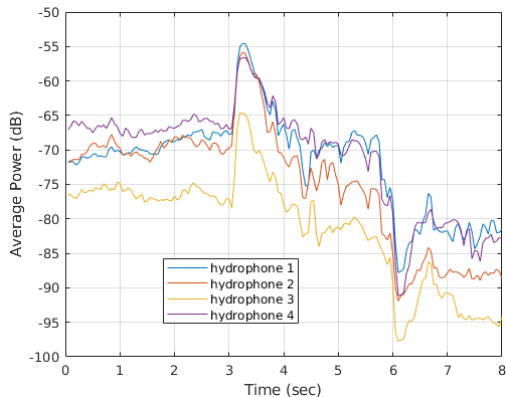
Introduction

- Signals can be analyzed and understood using various mathematical and statistical models
- Signal processing involves converting or transforming data in a way that allows us to see things in them that are not possible through direct observation
 - Signal processing has been used to understand the human brain, diseases, audio and image enhancements, financial market trends, and much more
- As mentioned, real world signals are not always suitable for use in ML models
 - Signals need to be processed so that they are cleaned, segmented, transformed, labeled, and fit to ML models
- Therefore, signal processing has become the mainstream of data analysis using ML models



Outline

- 1 Introduction
- 2 Machine Learning
- 3 Signal Processing
- 4 Using Signal Processing in ML
- 5 Application Examples
- 6 Conclusions



Machine Learning

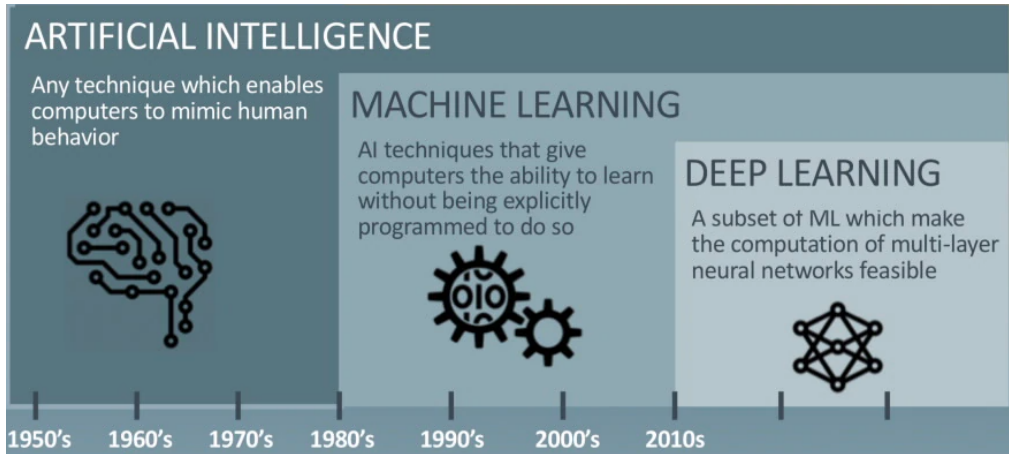


Artificial Intelligence

- Artificial Intelligence (AI) is the intelligence demonstrated by machines
 - Term coined in 1955 by John McCarthy, Alan Turing, Marvin Minsky, Allen Newell and Herbert Simon
- Today, AI is considered the simulation of intelligent/rational behavior
- Computational Intelligence: a computer or system designed to perceive its environment, understand its behavior, and take some action
 - Its main pillars are ML, Fuzzy Systems and Evolutionary Computation
- AI is estimated to create \$15.7 trillion in economic value worldwide by 2030, according to (2018) PwC forecasts
 - AI is transforming engineering in almost every industry and application area



Artificial Intelligence

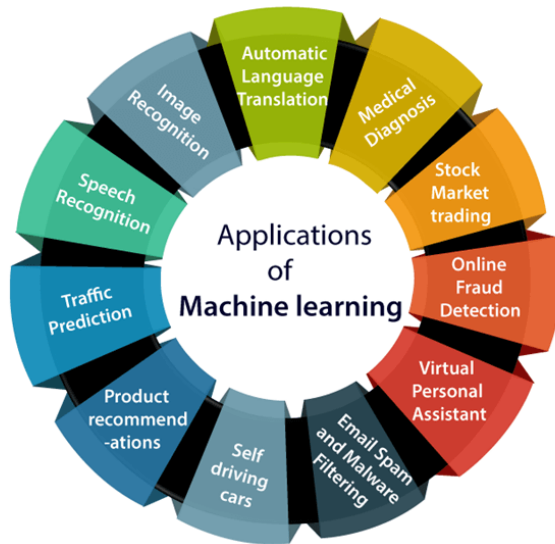


Machine Learning

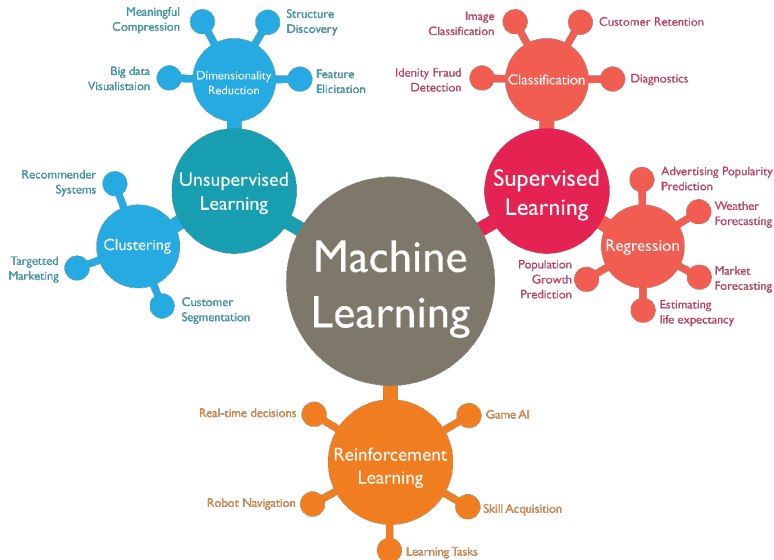
- ML teaches computers to do what comes naturally to humans and animals: *learn from experience*
- ML algorithms use computational methods to learn information directly from data without relying on a predetermined equation as a model
- ML algorithms find natural patterns in data that generate insight and help you make better decisions and predictions
- The algorithms adaptively improve their performance as the number of samples available for learning increases
- ML is currently used to build models for many engineering, scientific and social areas



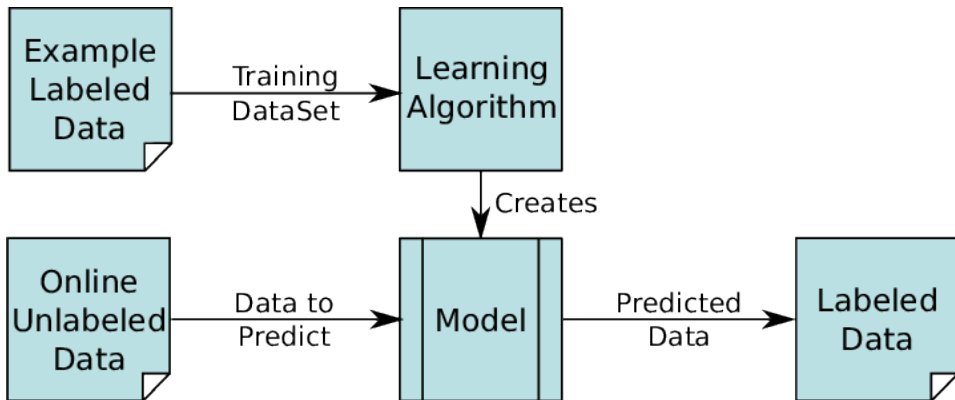
Machine Learning Applications



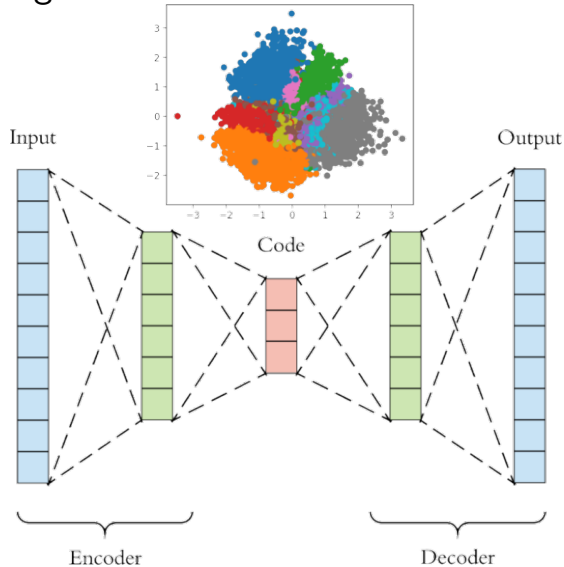
Machine Learning Algorithms



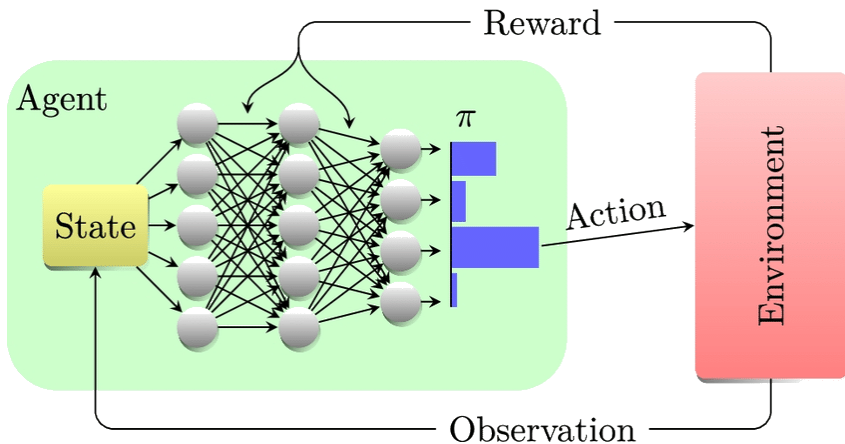
Supervised Learning



Unsupervised Learning



Reinforcement Learning



Deep Learning

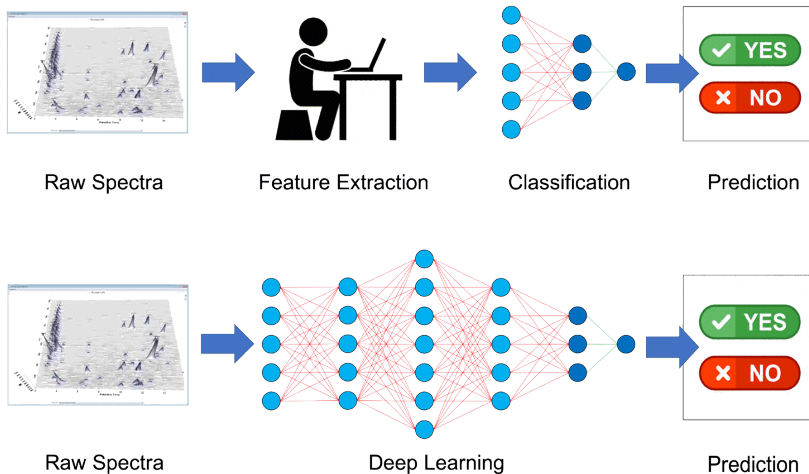


Deep Learning

- Deep Learning (DL) is a type of ML in which a model learns to perform tasks to solve real-world problems, directly from signals (*i.e.*, images, audio, text)
- DL is generally implemented using a *complex* neural network architecture
 - Traditional neural networks contain a couple of hidden layers, while DL networks can have hundreds of layers
 - Specifically, the term *deep* refers to the number of layers in the network – the more layers, the deeper the network
- Currently, DL is a standard and flexible technique that is evolving very fast
- DL is especially suitable for identification, speech recognition, text translation, advanced driver assistance systems and generative AI applications
 - DL is the state-of-the-art in classifiers because its accuracy



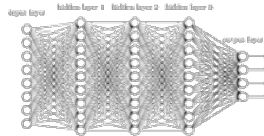
Deep Learning Convenience



Deep Learning Algorithms

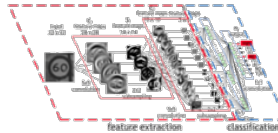
providing lift for
classification and
forecasting models

*Deep
Neural
Networks*



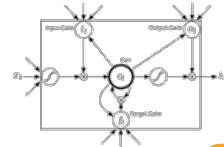
feature extraction
and classification of
images

*Convolutional
Neural
Networks*



for sequence of events,
language models, time
series, etc.

*Recurrent
Neural
Networks*



Signal Processing

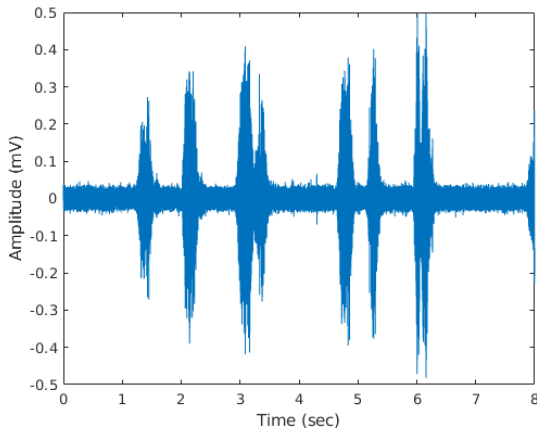


One-dimensional Processing



Motivation

- Let us start with a simple application of ML: identifying birds through their songs

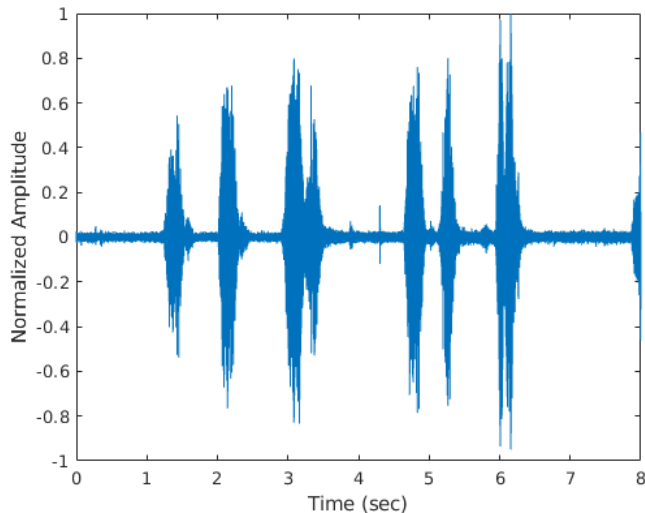


Time-domain Processing

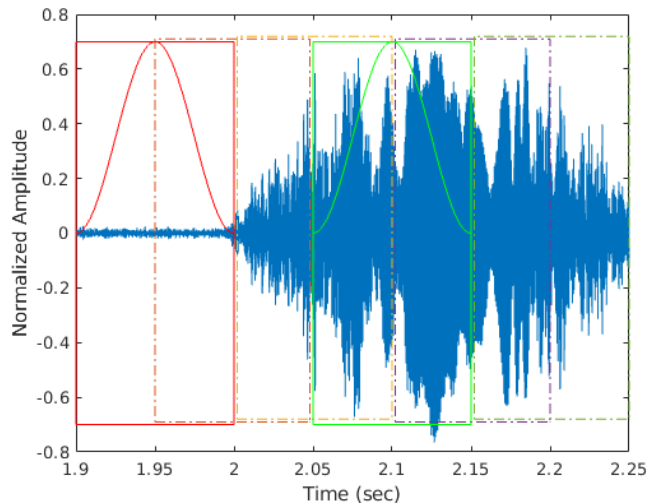
- Time-domain convolution
 - Filtering of the signal using a defined impulse response (mask)
 - Types of filters: low-pass, high-pass, band-pass, band-stop (notch)
 - Design considerations: FIR vs IIR filters
 - Adaptability considerations: Static vs adaptive filters
- Window functions (Segmentation)
 - Selective treatment usually requires apodization or tapering of the signals
 - This operation consists of multiplying the signal by a window of finite length whose amplitude varies smoothly and gradually towards zero at the edges
 - Type of windows: rectangular, triangular, Hanning, Hamming, Blackman, Kaiser



Time-domain Convolution



Time-domain Windows



Time-domain Characterization

- Signals can be characterized by some specific parameters in the time domain
- The statistical description produces the most basic features
 - These metrics are obtained considering the signal as a random process
 - Mean value, standard deviation, maximum value, minimum value, kurtosis, skewness, entropy, short-time energy, zero-crossing rate, and so on
- Linear prediction coefficients estimate the future values of a signal as a linear function of previous samples
- Correlation functions can also be used to describe signals in the time domain
 - The auto-correlation and cross-correlation functions give us similarity metrics between signals

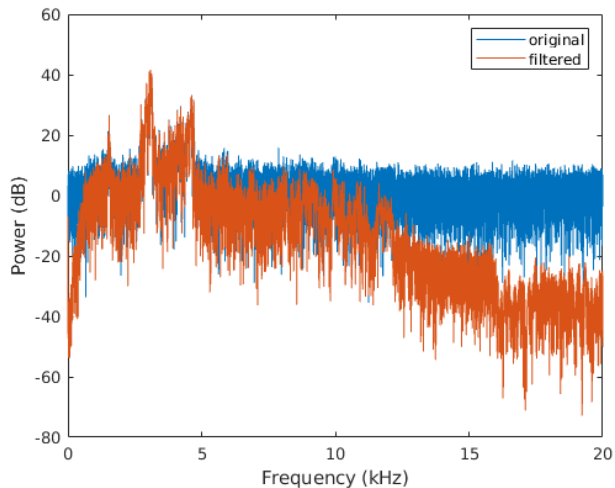


Frequency-domain Processing

- It usually starts with the Fourier transform of the windowed time signal (STFT)
 - It is a complex-valued decomposition that represents the out-of-phase sinusoids that constitute the original signal
 - Spectral components are generally best represented by their power spectral density
- There are also some useful frequency-domain statistical descriptors
 - Spectral centroid, spectral spread, spectral crest, spectral kurtosis, spectral skewness, spectral flatness, spectral entropy, and so on
- Filter operations are simpler than in the time domain
 - We have multiplication operations instead of convolutions
 - Very efficient operation for the implementation of filter banks



Frequency-domain Processing

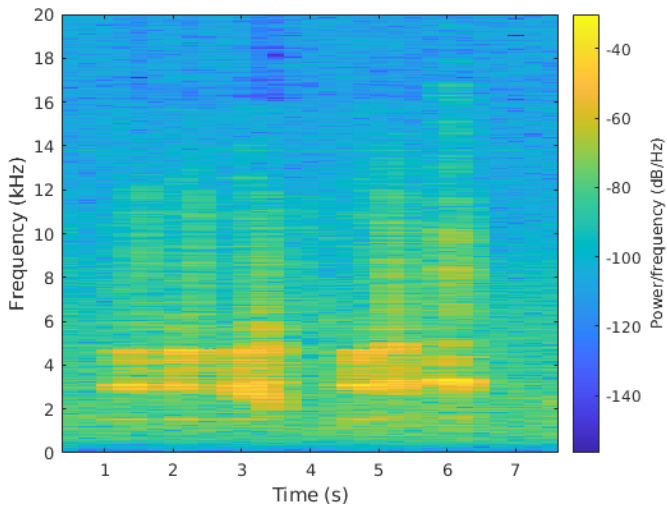


Time-Frequency Domain

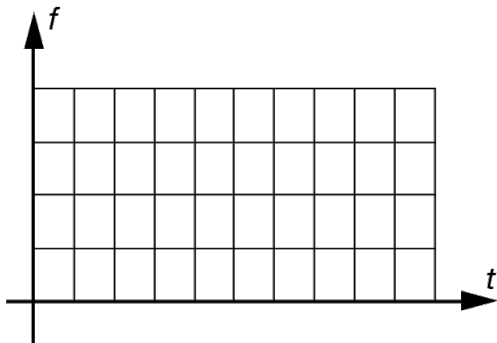
- There are also various time–frequency representations for signal analysis
- The most common visual representation is the spectrogram of a signal
 - It is usually depicted as a heat map
- There are other transforms for time-frequency analysis based on
 - Cosine and sine transforms
 - Mel frequency cepstral coefficients
 - Wavelet transform
- The main difference among the diverse representations is the efficiency to access localized information about the signal



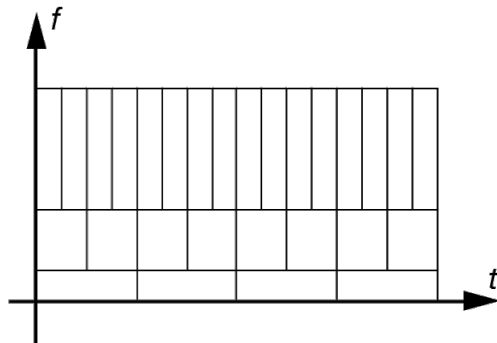
Time-Frequency Domain



Time-Frequency Domain



STFT



Wavelets

Multidimensional Processing



Multidimensional Processing

- The most typical example of multidimensional signal processing is digital image processing (DIP)
 - DIP has many advantages over analog image processing
 - DIP can be modeled as the operation of a multidimensional system
- Image processing can be performed in the *space* or frequency domain
- The main operations required by image processing are
 - Image enhancement
 - Image restoration
 - Image transformation
 - Image segmentation
 - Image compression



Image Enhancement

- Image enhancement is the process of adjusting digital images so that the results are more suitable for display or further image analysis
- It can be done using basic transformations (*i.e.*, contrast stretching or histogram equalization), spatial convolution or frequency-domain filtering

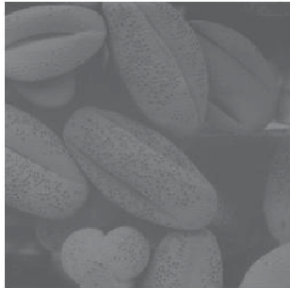


Image Restoration

- Image restoration is the operation of taking a corrupt/noisy image and estimating the original clean image
- It can be done using adaptive (notch) filters or by applying inverse filtering based on the estimated degradation function



Image Transformation

- Affine transformation techniques are used to correct geometric distortions or deformities that occur with non-ideal camera angles
- These transformations are currently used as a *data augmentation* technique

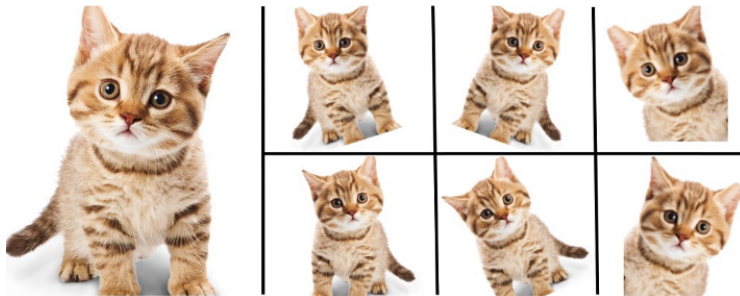


Image Segmentation and Compression

- Image segmentation is the process of partitioning an image into multiple segments, regions, or objects
 - It can be based on color and/or boundaries, and typically uses morphological operations to improve partitioning
- Image compression minimizes the size of the image in bytes without degrading image quality below an acceptable threshold
- Image segmentation and compression are high-level operations that are currently implemented using DL techniques
 - Convolutional autoencoders are used for semantic segmentation and image compression
 - Autoencoders are also used for image reconstruction, image coloring and other artistic transformations



Using Signal Processing in ML



Feature Extraction



Feature Extraction

- Feature extraction is a core activity in any traditional ML task
- Given a set of inputs, the goal is to discover a clear, informative, appropriate, and perhaps compact representation of those data
- Feature extraction involves two stages
 - Feature generation
 - Feature selection
- Feature generation requires the creation of new features that can improve the accuracy of the model
- Signal processing techniques are widely used for this purpose, although the entire process can be a bit more complex than expected



Feature Generation

- Data cleaning
 - Detect and correct (or remove) corrupt or inaccurate records from a dataset
 - Identify incomplete, incorrect, inaccurate or irrelevant parts of the data and then replace, modify, or delete the dirty or coarse data
- Data normalization
 - The goal is to transform the features so that they are on a similar scale
 - Common techniques are scaling to a range, z-score, clipping, and log-scaling
- Dimensionality reduction
 - Multiple transforms can exploit and remove information redundancies
 - Traditional techniques such as principal component analysis and singular value decomposition can be used
 - Currently, we can also use autoencoders for dimensionality reduction



Feature Selection

- Feature selection is an important step in an ML pipeline
- Feature selection is based on the importance of each feature for the classification or regression task to be performed
- Fortunately, there are some models that provide their own interpretation of the importance of features
- Lasso regression is one of such models
 - Lasso regression is a linear model that uses a cost function with an L1 penalty on the feature coefficients
 - Trying to minimize the cost function, the Lasso regression will automatically select those features that are useful, discarding the useless or redundant features



Data Augmentation



Data Augmentation

- Data augmentation is a technique for artificially increasing the training set by creating modified copies of a dataset using existing data
- Unlike synthetic data, augmented data is based on original data with some minor changes
 - Synthetic data is generated based on the original dataset using mainly Generative Adversarial Networks
- We use data augmentation to
 - Prevent models from overfitting
 - Improve the model accuracy and robustness
 - Reduce the operational cost of labeling and cleaning the raw dataset
 - Or when the initial training set is too small



Data Augmentation

- Augmentation techniques can be applied to audio, images, video, text, and other types of data as well
- Data augmentation can be done offline or online
- Finding an effective data augmentation approach can be very challenging
- We use common signal processing techniques for audio augmentation
 - Noise injection
 - Time shifting (by a random time)
 - Time stretching (*i.e.*, change the speed)
 - Random gain or polarity inversion
 - Pitch scaling and shifting
 - More advanced filtering operations in the frequency domain



Data Augmentation

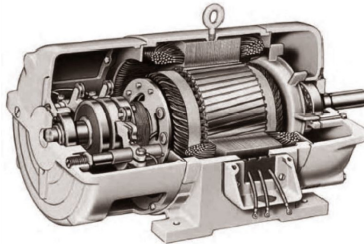
- Image augmentation also uses well-known image processing techniques
 - Geometric transformations: flip, crop, rotate, stretch, and zoom images randomly
 - Color space transformations: randomly change RGB color channels, contrast, and brightness
 - Kernel filters: randomly change the sharpness or blurring of the image
 - Random erasing: delete some part of the initial image
 - Mixing images: blending and mixing multiple images
- Other signals such as electrocardiograms, electroencephalograms, radar systems, seismo-volcanic events have their own characteristics
 - In all these cases, data augmentation must retain the intrinsic properties of the original signals

Application Examples



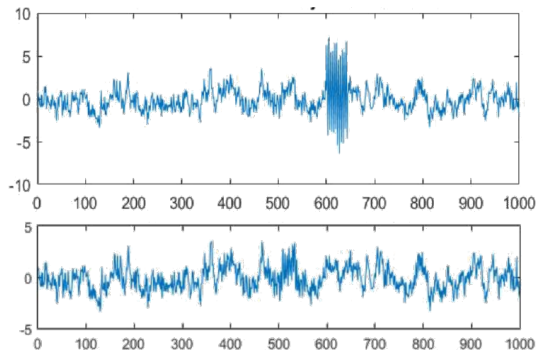
Electrical Signals

- Failure detection in induction motors using non-supervised ML algorithms
 - Toscano & Carrera (2020), *Advances in Intelligent Systems and Computing*, Vol. 1273, Ch. 5, Springer
 - Unsupervised learning for detecting failures as outliers (specificity = 99%)
 - Time and frequency features are clustered to find broken bars and faulty bearings



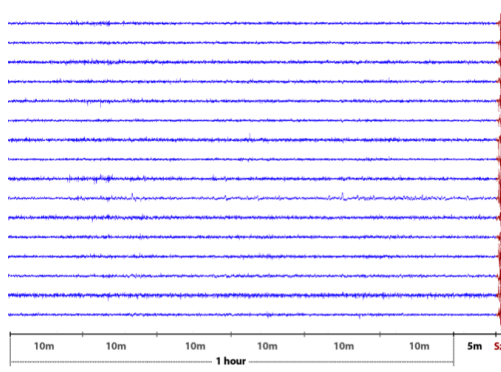
Radar Signals

- Target detection using radar processors based on machine learning
 - Carrera *et al.* (2020), *Proc. of Andescon'20*, IEEE
 - Traditional ML and DL techniques for detecting radar targets (accuracy $\sim 100\%$)



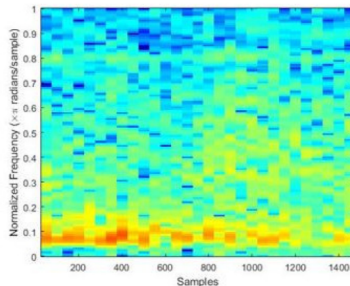
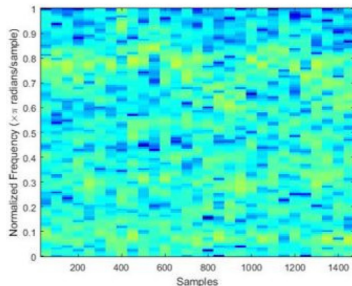
EEG Signals

- Analysis of epileptic seizure predictions based on intracranial EEG records
 - Carrera *et al.* (2018), *Proc. of ColCom'18*, IEEE
 - Traditional ML classifiers based on wavelet features (accuracy = 99%, 10 min)



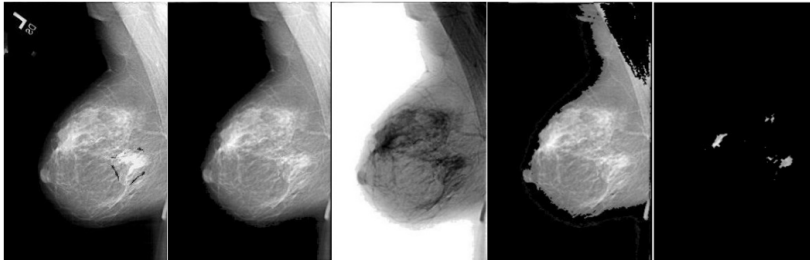
Seismo-volcanic Signals

- A deep learning approach for automatic recognition of seismo-volcanic events at the Cotopaxi volcano
 - Lara *et al.* (2021), *Journal of Volcanology and Geothermal Research*, Vol. 409, pp. 107142, Elsevier
 - Convolutional network based on transfer learning (accuracy = 99% for classification)



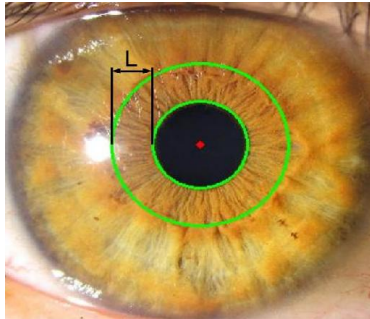
Mammogram Images

- Identification of breast cancer through digital processing of mammograms
 - Carrera, Sandoval & Carrasco (2021), *Communications in Computer and Information Science*, Vol. 1485, Ch. 30, Springer
 - Digital image processing for extracting a very small set of features (99% sensitivity)



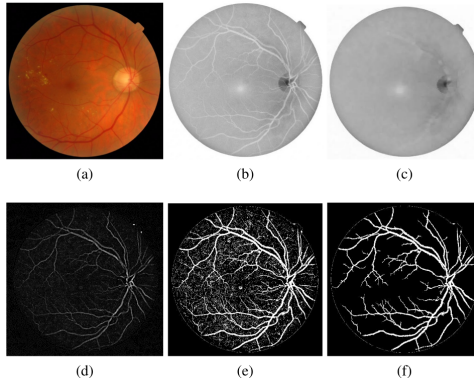
Iris Images

- Computer aided diagnosis of gastrointestinal diseases based on iridology
 - Carrera & Maya (2019), *Communications in Computer and Information Science*, Vol. 895, Ch. 40, Springer
 - Segmentation of the area of interest to get an accuracy of 94%



Fundus Images

- Automated detection of diabetic retinopathy using SVM
 - Carrera & Gonzalez (2017), *Proc. of Intercon'17*, IEEE
 - Detection of blood vessels, micro-aneurysms and hard exudates (sensitivity = 95%)



Conclusions



Conclusions

- AI is revolutionizing various fields of science and engineering by implementing autonomous decision-making systems
 - We look forward to the appearance of new applications in the coming years
 - Reinforcement and unsupervised learning are still in development
- Signal processing is a fundamental field in the areas of data analysis and ML
 - Time and frequency domain analyzes are essential for sample preparation and feature extraction
 - Image processing is a multidimensional processing very useful in convolutional neural networks
- DL does not eliminate the need to capture, transform and adapt most of the input samples



Conclusions

- Although autoencoders are a new alternative for feature extraction tasks, signal processing remains as the basic tool for sample preprocessing and adaptation
- Data augmentation is required to increase the size of the dataset for many DL applications
- The concepts discussed here have significant applications in engineering, medicine, security and defense, etc.
- Other successful works applying signal processing and ML are
 - Aided diagnosis system for skin cancer detection (2019), cancelable biometric schemes applied to iris recognition (2022), passive positioning of autonomous underwater vehicles (2022), finger-vein biometric system based on textural features (2018), recognizing different ventricular fibrillation types in low-dimensional latent spaces (2023), source coding for text transmission (2020), etc.



Thank You!



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